

MAC-Message Authentication Code

Alice and Bob share a key k

➔ HMAC - use a hash function on the message and the key

$$\text{MAC}_k(m) = H(k \parallel m)$$







$\min_{\mathbf{N}} \text{MAE}_k(\mathbf{n})$

MACKey



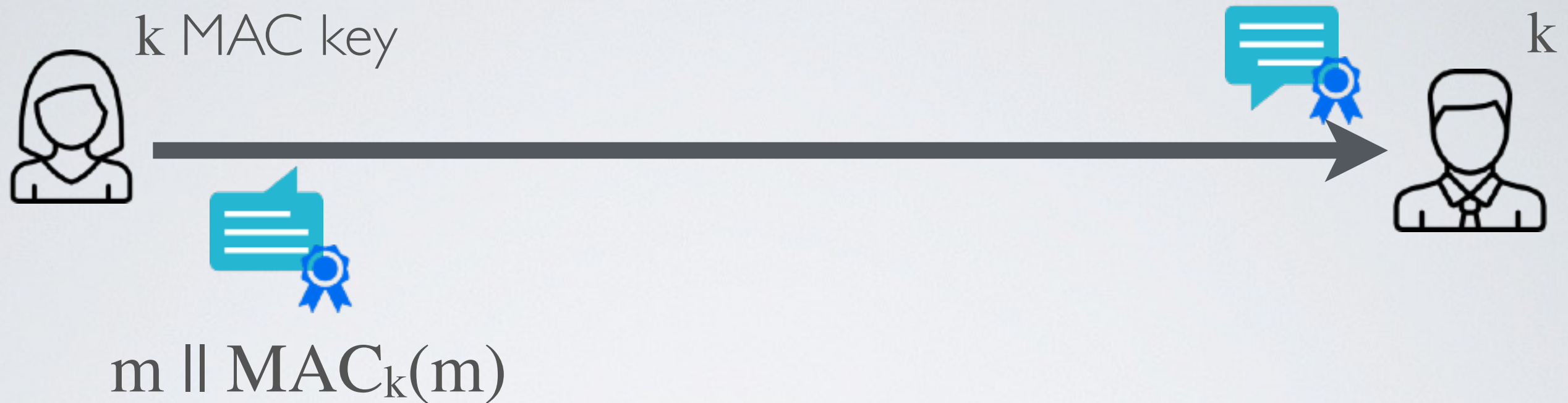








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Length Extension Attack

Vulnerable : All Merkle–Damgård-based hash functions
so MD5, SHA-1 and SHA-2 (but not SHA-3)